Computer Networks LAB (CS3P004)
Spring Semester 2024-25
Lab Sheet 1A

Ans – 1)

a) **ping** uses the ICMP protocol's mandatory ECHO_REQUEST datagram to elicit an ICMP ECHO_RESPONSE from a host or gateway. ECHO_REQUEST datagrams
pings have an IP and ICMP header, followed by a struct timeval and then an arbitrary number of "pad" bytes used to fill out the packet.

ping works with both IPv4 and IPv6. Using only one of them explicitly can be enforced by specifying -4 or -6.

Utility – Upon entering **ping** followed by a address, it returns the ping for that url, along with it's IP and packets received and packet loss and also the total time taken.

b) Below is the output upon enter the command: " **ping** 10.10.152.1 "

```
cslab2@cs-HP-ProOne-440-AIO:~$ ping 10.10.152.1
PING 10.10.152.1 (10.10.152.1) 56(84) bytes of data.
64 bytes from 10.10.152.1: icmp_seq=1 ttl=255 time=7.35 ms
64 bytes from 10.10.152.1: icmp_seq=2 ttl=255 time=11.1 ms
64 bytes from 10.10.152.1: icmp_seq=3 ttl=255 time=9.07 ms
64 bytes from 10.10.152.1: icmp_seq=4 ttl=255 time=8.36 ms
64 bytes from 10.10.152.1: icmp_seq=5 ttl=255 time=12.8 ms
64 bytes from 10.10.152.1: icmp_seq=6 ttl=255 time=7.94 ms
64 bytes from 10.10.152.1: icmp_seq=7 ttl=255 time=16.0 ms
64 bytes from 10.10.152.1: icmp_seq=8 ttl=255 time=10.5 ms
64 bytes from 10.10.152.1: icmp_seq=9 ttl=255 time=12.8 ms
64 bytes from 10.10.152.1: icmp_seq=10 ttl=255 time=8.91 ms
^C
--- 10.10.152.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 7.350/10.491/16.047/2.594 ms
```

We observe that the ping 10.10.152.1 i.e. our college IP is very low averaging around 10ms, which suggests a very high speed network connection.

c) Usage – **ping** command primarily works by using the format ping __"ip"__ . Additionally, we can give other options along with it like:

i. ping "ip" -4 returns the ping for IPv4, and similarly -6 returns the same for Ipv6.
ii. -a gives audible ping.
iii. -b allows pinging a broadcast address.

d) Results for "ping google.com" and "ping youtube.com" :

```
cslab2@cs-HP-ProOne-440-AIO:~$ ping google.com
PING google.com (142.250.192.142) 56(84) bytes of data.
64 bytes from bom12s18-in-f14.1e100.net (142.250.192.142): icmp_seq=1 ttl=114 time=58.6 ms
64 bytes from bom12s18-in-f14.1e100.net (142.250.192.142): icmp_seq=2 ttl=114 time=60.3 ms
64 bytes from bom12s18-in-f14.1e100.net (142.250.192.142): icmp_seq=3 ttl=114 time=60.5 ms
64 bytes from bom12s18-in-f14.1e100.net (142.250.192.142): icmp_seq=4 ttl=114 time=59.5 ms
64 bytes from bom12s18-in-f14.1e100.net (142.250.192.142): icmp_seq=5 ttl=114 time=60.0 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 58.555/59.787/60.501/0.696 ms
cslab2@cs-HP-ProOne-440-AIO:~$ ping youtube.com
PING youtube.com (142.251.42.14) 56(84) bytes of data.
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=1 ttl=113 time=60.2 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=2 ttl=113 time=66.2 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=3 ttl=113 time=63.0 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=4 ttl=113 time=64.4 ms
64 bytes from bom12s19-in-f14.1e100.net (142.251.42.14): icmp_seq=5 ttl=113 time=64.1 ms
^C
--- youtube.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 60.164/63.578/66.223/1.997 ms
```

Result for ping "iitbbs.ac.in" :

```
cslab2@cs-HP-ProOne-440-AIO:~$ ping iitbbs.ac.in
PING iitbbs.ac.in (10.10.0.26) 56(84) bytes of data.
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=1 ttl=63 time=4.55 ms
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=2 ttl=63 time=7.70 ms
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=3 ttl=63 time=7.01 ms
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=4 ttl=63 time=6.79 ms
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=5 ttl=63 time=6.55 ms
64 bytes from 10.10.0.26 (10.10.0.26): icmp_seq=6 ttl=63 time=9.12 ms
^C
--- iitbbs.ac.in ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5007ms
rtt min/avg/max/mdev = 4.549/6.953/9.123/1.368 ms
```

Differences:- We observe that external websites have a ping of around 60-65ms. On the other hand the ping of our college address is very low, the reason for this is we are using a Wi-Fi connected

to SES and local network allows it to be much faster almost 10 times than the others.

Ans – 2)

a) **ifconfig** required "net-tools" to be externally installed apart from build-essential".
Description - ifconfig is used to configure the kernel-resident network interfaces. It is used at boot time to set up interfaces as necessary. After that, it is usually only needed when debugging or when system tuning is needed.
Other than that it displays the status of currently active interfaces of a single or multiple interfaces, based on the arguments given.

b)

```
cslab2@cs-HP-ProOne-440-AIO:~$ ifconfig
eno1: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether 28:c5:c8:bd:c1:01  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 19  memory 0x80900000-80920000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 847895  bytes 60713729 (60.7 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 847895  bytes 60713729 (60.7 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlp0s20f3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.10.68.164  netmask 255.255.252.0  broadcast 10.10.71.255
        inet6 fe80::1d69:2d4b:fce:ca5b  prefixlen 64  scopeid 0x20<link>
        ether 5c:b4:7e:86:ea:89  txqueuelen 1000  (Ethernet)
        RX packets 737454  bytes 947302987 (947.3 MB)
        RX errors 0  dropped 55  overruns 0  frame 0
        TX packets 126404  bytes 16273503 (16.2 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

i. IPv4 of my computer – 10.10.68.164
ii. MAC address – 5c : b4 : 7e : 86 : ea : 89
iii. **Loopback interface** – a software-only interface that emulates the physical interface. It can be used to identify a device and can be used to determine if the device is online.

Also, the loopback interface never changes and is always up.

iv. **IPv4** of loopback interface – 127.0.0.1

c) **up –** causes the interface to be activated
down – It is opposite of **up** and causes the driver for that interface to shut down.

Ans – 3)

a) A domain name gets converted to its ip-address using the DNS – *Domain Name System.*
b) **nslookup - nslookup** is a program to query Internet domain name servers.
nslookup has two modes: interactive and non-interactive.
Interactive mode allows the user to query name servers for information about various hosts and domains or to print a list of hosts in a domain.
Non-interactive mode prints just the name and requested information for a host or domain.
c) **IPv4** of iitbbs.ac.in – 14.139.195.4 and 14.139.204.220
secs.iitbbs.ac.in – 10.10.0.172
google.com – 142.250.183.78
facebook.com – 157.240.16.35
youtube.com – 142.251.42.110
IPv6 of google.com – 2404 : 6800 : 4009: 822 : : 200e
youtube.com – 2404 : 6800 : 4009 : 832 : : 200e
d) Yes, a website can have more than one IP address. Yes, one example was the domain name iitbbs.ac.in having 2 IP's.
The reason for this is to reduce load the other one when one of them is too heavy.

Ans – 4)

a) **traceroute** command is used to track the path.
b) Utility - **traceroute** tracks the route packets taken from an IP network on their way to a given host.
It utilizes the IP protocol's time to live (TTL) field and attempts to elicit an ICMP TIME_EXCEEDED response from each gateway along the path to the host.
c)
i. We observe that each of them has max hop = 30 and packet size = 60 bytes.
ii. **traceroute** to onlinesbi takes so long because of high server load or possible maintainance going on in the website.

iii. The first hop is the same in all of them because that's usually the home router or in this case the iitbbs network. Also, this might be due to target firewall blocking incoming requests, so as to block possible DDOS attack.
iv. * * * - represents that it caused a timeout and unable to reach the server for that hop.
v. Time timestamps at each point on the traceroute represents the time in required to get a response from the server for each hop/step along the route.
vi. We can locate at which point the packet leaves the campus network and the ISP by looking at the Timeouts (* * *) and the latency of requests.
The increasing latency at the beginning suggests the packet going further from the campus and eventually going beyond. Finally * * * possibly represents it leaving the campus.
Then in the middle the latency is constant reprensenting it in the ISP network and then finally * * * repreesnting it leaving ISP.
Also, we can look at the IP changes to detect where the packet is currently at the first 3 digits of IPv4 represents the network remaining the same.

Ans – 5)

a) **scp** copies files between hosts on a network.
It uses ssh(1) for data transfer, and uses the same authentication and provides the same security as a login session.
scp will ask for passwords or passphrases if they are needed for authentication.
Utility – It can be used to share files within a network.